CS 4782 Coding Assignment 2 Written Responses

Due: 3/6/25 11:59 PM on Gradescope

Late submissions accepted until 3/8/25 11:59 PM

Note: For homework, you can work in teams of up to 2 people. Please include your teammates' NetIDs and names on the front page and form a group on Gradescope. (Please answer all questions within each paragraph.) Please show all the relevant steps in your solutions.

Question 1:

Based on the provided implementation of *PositionalEncoding*, what is the formula used to assign an embedding to each position/index i in the input sequence? What is one benefit of using this function/formula specifically to generate position embeddings?

The formula as a function of position i and embedding dimension j where all even dimensions are defined as:

$$\mathrm{PE}(i, j) = \begin{cases} \sin\left(i/10000^{j/d_{\text{model}}}\right) & \text{if } j \text{ is even} \\ \cos\left(i/10000^{(j-1)/d_{\text{model}}}\right) & \text{if } j \text{ is odd} \end{cases}$$

A benefit of this sinusoidal encoding is that it allows the model to generalize to input lengths longer than those that have been seen in training because the relative position between any two tokens can be writen through a linear combination of the exisiting encoding values.

Question 2:

Describe the matrix size transformations as they happen from steps 1-5. Specificaly, what is the shape of:

1. the input to the **split heads** function?

   (b, n, d-model)

2. the output from the **split heads** function?

   Q: (b, h, n, d-q) K: (b, h, n, d-k) V: (b, h, n, d-v)

3. **multi-head** variable in the description above?

   (b, n, d-model)

4. final output of **Multi-Head(Q, K, V)**?

   (b, n, d-model)

Answers expected as matrix shapes in terms of n, d_{model}, d_q, d_k, d_v and h. You can use the additional variable b to represent batch size.

Question 3:

What do you notice about the graph above and the differences between the five models you have implemented? Are the results consistent with what you expected? How do pre-trained models compare to models you trained from scratch? Write 3-4 sentences below.

The most impressive performance increase is when we apply word2vect to logistic regression. I find it odd that LSTM and LR + W2V have such similar performance. distilBERT is the only truely pretrained model here and it provides significant better test accuracy over all other models at this task. Additional, W2V could be considered pretrained and made a massive improvement over the base logistic regression model.